AI-Driven Integrated Healthcare Monitoring and Disease Risk Prediction Using Wearable Sensors and Machine Learning

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***Abstract*—Continuous wrist photoplethysmography is prone to motion artifacts, optical decoupling, and telemetry drops that trigger spurious resting tachycardia alarms. While digital health platforms routinely deploy multi-signal software quality metrics to suppress false alarms, the incremental utility of software trust scoring over simple kinematic motion gating is rarely benchmarked under controlled protocols. We present a reproducible benchmark evaluating six causal alerting arms across fifteen ambulatory subjects from the public PPG-DaLiA dataset under a pre-specified, version-tagged protocol without an external public registry. To address natural tachycardia sparsity in healthy individuals, labelled synthetic resting tachycardia episodes were injected into derived pulse-rate streams alongside two seeded telemetry fault regimes and high-motion optical noise. Pre-specified hypothesis estimation found that kinematic stillness gating accounted for the largest share of false-alert suppression, whereas multi-attribute software trust scoring yielded no statistically significant incremental benefit across any evaluated condition. The no-motion trust ablation alone reduced false alerts by about half relative to personal baselines. Furthermore, conservative software abstention degraded event sensitivity under optical noise (exploratory, driven by two subjects). Inductive conformal abstention, pre-specified in Amendment 3, implemented after the other arms' results were seen, achieved the lowest false-alert frequency at the expense of recall. All findings are derived from synthetic episodes in healthy volunteers and carry no clinical standing.**

***Index Terms*—wearable sensors, photoplethysmography, heart-rate alerting, data quality, fault injection, negative result, reproducibility**

I. Introduction

Ambulatory continuous monitoring with consumer and medical-grade wearables offers substantial potential for early arrhythmia detection. However, wrist photoplethysmography (PPG) is acutely sensitive to motion artifacts, optical decoupling, transmission dropouts, and flatline sensor freezes. When raw optical or derived telemetry signals degrade, alerting algorithms frequently emit spurious notifications. In wearable monitoring and clinical telemetry alike, frequent false alerts contribute to monitor alarm fatigue [12].

To mitigate spurious alarms, modern digital health architectures deploy signal quality indices (SQIs) and multi-component trust heuristics that penalize missing samples, flag frozen values, and bound rate-of-change dynamics. Despite their widespread adoption, the isolated incremental utility of multi-component software trust scores over straightforward kinematic stillness gating (e.g., Euclidean Norm Minus One, ENMO) has rarely been benchmarked under controlled cross-validation. The core protocol was version-tagged before the alerting experiments; later amendments are listed in Table VII; no external public registry was used.

This paper reports a pre-specified fault-injection benchmark for wearable heart-rate alerting. Our contributions are:

1) **Open Benchmarking Framework:** A reproducible pipeline formalizing seven fault types plus a clean condition (gaps, stale values, duplicates, spikes, drift, scaling errors, and raw optical noise) evaluated across 35.94 total recording hours (31.27 non-event exposure hours) of multi-activity ambulatory recordings.
2) **Causal Leave-One-Subject-Out (LOSO) Evaluation:** Six distinct alerting arms, tested under rigorous data quarantine, separating training folds from held-out evaluation subjects.
3) **Rigorous Negative Finding:** Empirical evaluation of pre-specified estimation hypothesis H_1, showing that kinematic motion gating provides a 65.9%–74.2% reduction in false alerts relative to the ungated personal baseline, while multi-attribute software trust scoring provides no statistically detectable incremental benefit ($p \geq 0.28$).

II. Literature Survey

A. Motion Artifact Mitigation in Wrist PPG: Wrist photoplethysmography (PPG) measures the pulsatile changes in blood volume in the cutaneous microvascular bed optically using illuminated LEDs and paired photodiodes [3].

In ambulatory environments, physical movement produces motion artifacts that interfere with optical pulse monitoring [7]. These motion-induced spectral peaks directly overlap cardiac fundamental frequencies (1.0–3.5 Hz, or 60–210 bpm), biasing unconstrained peak detection and Fourier pulse-rate estimates. Wearable signal processing for recovering corrupted cardiac signals has come to rely on multi-channel kinematic fusion. Zhang *et al.* [8] proposed the TROIKA framework, which integrates temporal difference filtering, sparse signal reconstruction and peak tracking against triaxial accelerometry with 12 treadmill subjects. Reiss *et al.* [1] introduced the 15-subject PPG-DaLiA benchmark hosted on the UCI repository [5] together with deep convolutional tracking architectures, establishing standard Mean Absolute Error (MAE) baselines against chest ECG and identifying performance bottlenecks during non-rhythmic wrist movements. Later wearable telemetry corpora, e.g. the PhysioNet Wearable dataset [6] have extended the stress and multi-activity benchmark protocols. In addition, Bent *et al.* [7] studied sources of inaccuracy in wearable optical sensors and identified motion artifacts for different types of physical activity.

B. Signal Quality Indices and Multi-Modal Fusion: Automated Signal Quality Indices (SQIs) for discarding unreliable epochs have been widely studied [2], [3] as signal reconstruction cannot completely recover severely decoupled waveforms. Elgendi [2] investigated eight different morphological and statistical SQIs, emphasizing skewness, kurtosis, and perfusion indices on stationary recordings. Signal quality estimation was formulated by Karlen *et al.* [9] using repeated Gaussian filters and cross-correlation on photoplethysmogram waveforms. Cross channel telemetry fusion for improving alert fidelity in wireless patient monitoring. Orphanidou *et al.* [10] developed unified ECG and PPG quality metrics by combining envelope analysis, template matching and physiological bounds, and validated utility on ambulatory wireless recordings. Li and Clifford [11] evaluated signal quality indices and data fusion for false alarm reduction in intensive care telemetry.

C. Alarm Fatigue, Telemetry, and Conformal Uncertainty: Frequent false alerts from bedside and wearable monitors incur large clinical and behavioral costs. In hospital settings with high false alarm rates desensitize clinicians, and monitor alarm fatigue was reviewed by Cvach [12]. This problem was formalized by Clifford *et al.* [13] in the PhysioNet/CinC 2015 Benchmark ($N = 1250$ multi-parameter records) to tackle false alarm suppression with multi-modal physiological signals. Deterministic predictions are not distribution-free confidence under non-stationary ambulatory noise. Conformal prediction [14] offers finite-sample, distribution-free coverage guarantees with valid non-conformity calibration. Vazquez and Facelli [4] reviewed conformal prediction in fourteen clinical applications for quantifying predictive uncertainty in medical settings. Table I compares this benchmark with literature sources opened and verified.

III. Dataset and Protocol

A. PPG-DaLiA Dataset

We evaluate on the public PPG-DaLiA benchmark (UCI ID 495, CC BY 4.0) [1], which contains 15 healthy subjects performing 2.5 hours of authentic daily activities (Table II). Gold-standard chest ECG is used to derive reference heart rates. Raw wrist PPG (64 Hz) and triaxial accelerometry (32 Hz) were recorded with an Empatica E4 wristband. Strict quarantine was maintained: chest ECG reference was isolated solely for ground truth evaluation and was inaccessible to alerting arms.

B. Labelled Synthetic Tachycardia Injection

Since natural resting tachycardia was virtually absent in this healthy volunteer cohort (manifesting in only 3 of 15 subjects), evaluating sensitivity on raw data is statistically uninformative. According to Amendment 2, we injected 112 labeled synthetic resting tachycardia episodes into derived pulse-rate streams during verified low-motion windows ($\text{ENMO}_{10\text{s}} \leq 0.05$ g). Peak heart rates sampled from 105 to 135 bpm, linear ramp durations of 15 s and minimum inter-episode gaps of 60 s. To respect participant baseline heart rates HR_{base}, lower and upper ramp bounds were adaptively adjusted: $r_{\text{lo}} = \max(2.0, (105 - HR_{\text{base}})/15)$, $r_{\text{hi}} = \min(4.0, (135 - HR_{\text{base}})/15)$, with plateau peak heart rate clamped to $[105.0, 135.0]$ bpm.

IV. System Architecture

The experimental evaluation harness and replay platform were implemented as a modular software suite for systematic wearable sensor benchmarking (Fig. 1). The tooling couples automated telemetry replay with the `vitasense.quality` multi-attribute trust engine and precomputed cross-validation execution caches.

V. Methods and Algorithms

A. Evaluated Alerting Arms and Mathematical Formulations

We benchmark six causal alerting arms without future peeking: Arm 1 (Fixed Population), Arm 2 (Personal Baseline), Arm 3a (+Kinematic Gating), Arm 3b (+Trust Score), Arm 3b-noMotion (Trust Ablation), and Arm 4 (Conformal Abstention). Decision rules, thresholds, and tuning grids are in Table III.

Causal Baseline and Shrinkage (Arms 2–4): The personal alert threshold $\tau_i(t)$ is dynamically tracked over causal stillness windows ($\text{ENMO}_{10\text{s}} \leq \theta_{\text{cal}}$, fixed at $\theta_{\text{cal}} = 0.05$ g) as:

$$\tau_i(t) = \max\left(\tau_{\text{floor}},\, \hat{\mu}_i(t) + k \cdot \hat{\sigma}_i(t)\right) \tag{1}$$

where empirical Bayes shrinkage regularizes running subject moments $(\bar{x}_i(t), s_i(t))$ toward population priors $(\mu_{\text{pop}}, \sigma_{\text{pop}})$:

$$w(t) = \frac{n(t)\Delta t/3600}{n(t)\Delta t/3600 + \kappa} \tag{2}$$

$$\hat{\mu}_i(t) = w(t)\bar{x}_i(t) + (1 - w(t))\mu_{\text{pop}} \tag{3}$$

$$\hat{\sigma}_i(t) = w(t)s_i(t) + (1 - w(t))\sigma_{\text{pop}} \tag{4}$$

TABLE I
RELATED WORK COMPARISON ON VERIFIED LITERATURE

Study	Data	Method	Quality / Uncertainty	Focus / Finding
Zhang *et al.* [8]	12 subjects	Sparse recon.	ACC spectrum	Exercise motion tracking
Reiss *et al.* [1]	15 subjects	Deep CNN	None	Daily living tracking benchmark
Elgendi [2]	Static PPG	Feature ranking	8 optical SQIs	Evaluation of optical indices
Charlton *et al.* [3]	Wearable lit.	Consensus review	SQI assessment	Wearable PPG roadmap
Clifford *et al.* [13]	1250 records	Multi-parameter	Multi-channel SQI	ICU false alarm challenge
Vazquez & Facelli [4]	14 studies	Conformal review	Non-conformity	Clinical decision confidence
This Benchmark	15 subjects	Causal threshold	Motion vs trust	Controlled motion vs trust benchmark

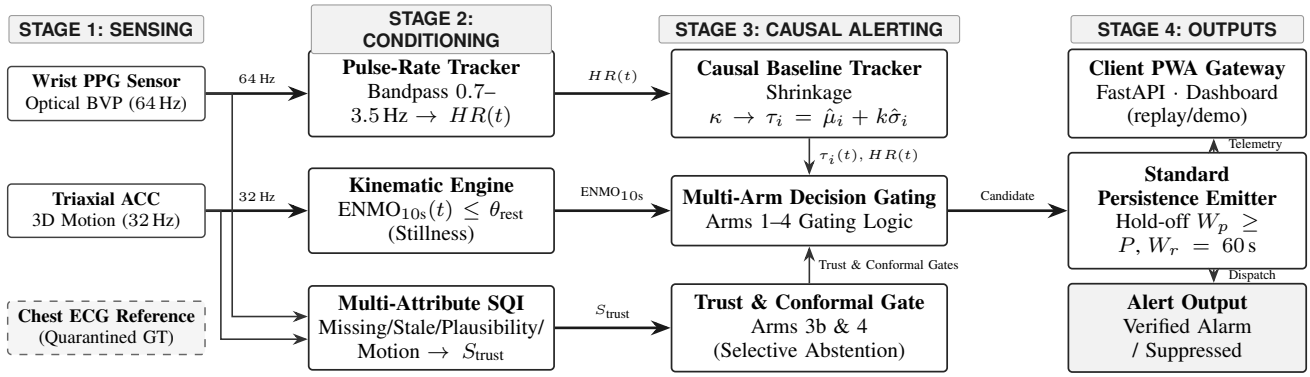


Fig. 1. System Architecture: End-to-end wearable telemetry, signal conditioning, causal baseline estimation, multi-arm gating and standard persistence emitter schematic.

TABLE II
PPG-DALIA BENCHMARK DATASET CHARACTERISTICS & EVALUATED FAULT REGIMES

Part A: PPG-DaLiA Benchmark Dataset Characteristics [1]	
Subjects	$N = 15$ healthy individuals (7 male, 8 female, age 21–55)
Duration	≈ 2.4 h per subject (35.94 total recording hours; 31.27 non-event exposure hours)
Activities	Sitting, Walking, Stairs, Cycling, Driving, Lunch, Soccer
Sensors	Wrist Empatica E4 (BVP: 64 Hz, 3D ACC: 32 Hz)
Ground Truth	Chest ECG (RespiBAN, 700 Hz, strictly quarantined)
Part B: Evaluated Telemetry & Sensor Fault Regimes (Amendment 3)	
Condition 0	Clean ambulatory background (31.27 h non-event exposure)
Regime A	Low-duty ($\approx 10\%$ faults): sporadic gaps (F1), freeze (F2), duplicates (F3), drift (F5), scale errors (F6)
Regime B	Nominal-duty ($\approx$ 25–30% faults): prolonged flatlines, frequent dropouts, scale errors (seed 42)
F7 Noise	Additive Gaussian noise continuously modulated by wrist 3D acceleration injected into raw BVP (SNR +5 dB)

with window count $n(t)$, step size $\Delta t = 2$ s, and shrinkage time constant $\kappa \in \{0.5, 1, 2\}$ h.

Multi-Attribute Signal Trust (Arms 3b & 4): Following `vitasense.quality.trust_score`, composite quality $S_{\text{trust}}(t) \in [0, 1]$ is evaluated as:

$$S_{\text{trust}}(t) = S_{\text{gap}}(t) \cdot S_{\text{stale}}(t) \cdot S_{\text{phys}}(t) \cdot S_{\text{mot}}(t) \tag{5}$$

penalizing missingness (S_{gap}), zero-variance flatlines (S_{stale}), unphysiological pulse limits (S_{phys}) and accelerometer dynamic motion (S_{mot}).

Conformal Prediction and Abstention Gate (Arm 4): Nonconformity calibration on stillness training windows evaluates $s(t) = |HR(t) - \hat{\mu}_i(t)|/\hat{\sigma}_i(t)$, producing $(1-\alpha)$ calibration quantile $q_{1-\alpha}$ and prediction band $[\hat{\mu}_i(t) \pm q_{1-\alpha}\hat{\sigma}_i(t)]$. Total augmented uncertainty is:

$$U_{\text{total}}(t) = \hat{\sigma}_i(t)/\hat{\mu}_i(t) + \lambda\,(1 - S_{\text{trust}}(t)) \tag{6}$$

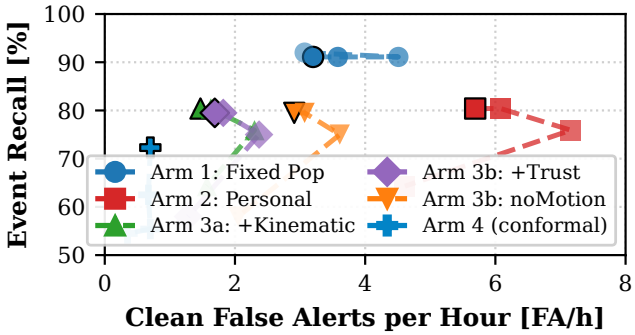
TABLE III
EVALUATED ALERTING ARMS, DECISION RULES, AND TUNING GRIDS (REFRACTORY 60 S FOR ALL ARMS)

Alerting Arm	Decision Rule & Tuning Grid
Arm 1: Fixed Pop.	$HR(t) > \tau_{\text{pop}}$; $\tau_{\text{pop}} \in \{100, 110, 120, 125, 130\}$ bpm, $P \in \{10, 20, 30\}$ s
Arm 2: Personal	$HR(t) > \max(\tau_{\text{flr}}, \hat{\mu}_i(t) + k \cdot \hat{\sigma}_i(t))$; $k \in \{2.0, \ldots, 4.0\}$, $\kappa \in \{0.5, 1, 2\}$ h, $P \in \{10, 20, 30\}$ s
Arm 3a: +Kinematic	Arm 2 $\wedge$ $(\text{ENMO}_{10s}(t) \le \theta_{\text{rest}})$; $\theta_{\text{rest}} \in \{0.03, 0.05, 0.08\}$ g, $P \in \{10, 20, 30\}$ s
Arm 3b: +Trust	Arm 2 $\wedge$ $(S_{\text{trust}}(t) \ge \theta_{\text{trust}})$; $\theta_{\text{trust}} \in \{0.60, 0.70, 0.80\}$, $P \in \{10, 20, 30\}$ s
Arm 3b-noMotion	Arm 2 $\wedge$ $(S_{\text{gap}} S_{\text{stale}} S_{\text{phys}} \ge \theta_{\text{trust}})$; $\theta_{\text{trust}} \in \{0.60, 0.70, 0.80\}$, $P \in \{10, 20, 30\}$ s
Arm 4: Conformal*	Arm 2 $\wedge$ $\neg$Abstain(t); $\alpha \in \{0.05, 0.10\}$, $\lambda \in \{0.5, 1.0\}$, $\theta_{\text{unc}} \in \{0.2, 0.3\}$
**Pre-specified in Amendment 3, implemented after other arms' results were seen.*	

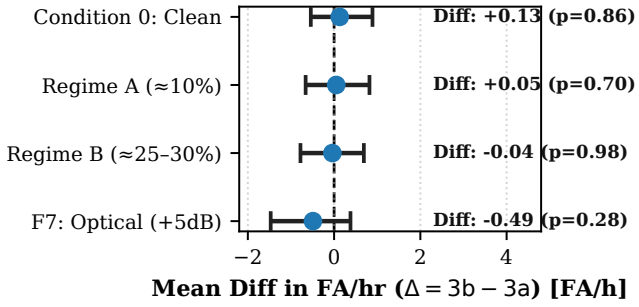
Arm 4 abstains (Abstain(t) = TRUE) if the prediction band spans threshold $\tau_i(t)$ ($[\hat{\mu}_i - q_{1-\alpha}\hat{\sigma}_i \le \tau_i(t) \le \hat{\mu}_i + q_{1-\alpha}\hat{\sigma}_i]$) or if $U_{\text{total}}(t) > \theta_{\text{unc}}$.

B. Shared Standard Alert Emitter

To ensure rigorous cross-arm comparability, all arms feed instantaneous binary triggers $C[t]$ into a shared `StandardAlertEmitter` (Algorithm 1). Alerts fire if and only if triggering conditions hold continuously for persistence window $W_p = P/\Delta t$ ($P \in \{10, 20, 30\}$ s). Emitted alerts enforce a mandatory refractory hold-off of $W_r = R/\Delta t$ ($R = 60$ s).



(a) Event Recall vs. Clean False Alerts/hr



(b) Paired Differences (Δ = Arm 3b − Arm 3a)

Fig. 2. Empirical benchmark findings across arms and conditions: (a) event recall versus clean false alerts per hour across evaluated arms and noise regimes, and (b) paired differences (Δ = Arm 3b − Arm 3a) with 95% bootstrap confidence intervals. Per-subject clean false alerts per hour are archived in `per_subject_metrics.csv`.

Algorithm 1 Shared Alert Emitter with Persistence and Refractory Hold-off

Require: Trigger stream $C[t]$, persistence window W_p, refractory window W_r

Initialize: $run_len \leftarrow 0$, $ref_cnt \leftarrow 0$
for each window $t = 0, 1, \ldots, N-1$ **do**
  if $ref_cnt > 0$ **then**
    $ref_cnt \leftarrow ref_cnt - 1$; $Alert[t] \leftarrow$ FALSE
  else if $C[t]$ is TRUE **then**
    $run_len \leftarrow run_len + 1$
    if $run_len \geq W_p$ **then**
      $Alert[t] \leftarrow$ TRUE; $ref_cnt \leftarrow W_r$; $run_len \leftarrow 0$
    else
      $Alert[t] \leftarrow$ FALSE
    end if
  else
    $run_len \leftarrow 0$; $Alert[t] \leftarrow$ FALSE
  end if
end for
return Alert stream $Alert$

TABLE IV
PAIRED STATISTICAL COMPARISON: ARM 3B VS ARM 3A

Condition	Mean Δ	95% Boot. CI	Wilcoxon W	p-val
Condition 0 (Clean)	+0.13	[-0.54, +0.89]	31.0	0.86
Regime A (≈10%)	+0.05	[-0.66, +0.82]	40.0	0.70
Regime B (≈25–30%)	-0.04	[-0.78, +0.69]	52.0	0.98
F7 (Optical Noise)	-0.49	[-1.47, +0.38]	17.0	0.28

TABLE V
STRATIFIED ALERTING METRICS (RECALL % / FA PER HOUR)

Condition	Stratum	Arm 1	Arm 2	Arm 3a	Arm 3b	3b-nM
Cond 0 (Clean)	All-15	91.1/3.2	80.4/5.7	80.4/1.5	79.5/1.7	79.5/2.9
	without S5	90.7/2.9	79.6/5.3	79.6/1.4	78.7/1.7	78.7/2.9
	S5/6/11 (3)	100.0/9.4	94.4/16.1	94.4/5.1	94.4/3.9	94.4/6.7
Reg A (≈10%)	All-15	91.1/3.6	80.4/6.1	80.4/1.6	79.5/1.8	79.5/3.1
	without S5	90.7/3.3	79.6/5.5	79.6/1.4	78.7/1.8	78.7/3.0
	S5/6/11 (3)	100.0/9.8	94.4/17.1	94.4/5.5	94.4/4.1	94.4/7.2
Reg B (≈25–30%)	All-15	91.1/4.5	75.9/7.2	75.9/2.3	75.0/2.4	75.0/3.6
	without S5	90.7/4.1	75.0/6.4	75.0/2.2	74.1/2.3	74.1/3.5
	S5/6/11 (3)	100.0/10.2	94.4/18.8	94.4/5.9	94.4/4.3	94.4/7.2
F7 (Optical)	All-15	92.0/3.1	64.3/4.6	64.3/1.6	58.0/1.3	58.0/2.0
	without S5	91.7/2.7	63.0/4.0	63.0/1.2	56.5/1.1	56.5/1.9
	S5/6/11 (3)	100.0/9.0	88.9/14.9	88.9/5.9	88.9/3.5	88.9/5.5

VI. EXPERIMENTAL SETUP

A. Leave-One-Subject-Out Cross-Validation & Scoring

Hyperparameters were tuned strictly within 15 LOSO training folds. For test subject $s \in \{1 \ldots 15\}$, candidate vectors were evaluated across the 14 training subjects pooled. The pre-specified rule selects the parameters that achieve $\geq 85\%$ training recall with the minimum false alerts per hour (FA/hr). If this is not achievable, the pre-specified fallback rule selects the setting that maximizes training recall.

Scoring assesses event-level recall (injected episodes within $[t_{\text{start}}, t_{\text{end}}]$) alongside alert-level clean false alerts per hour (FA/hr), precision, and F_1 score. Paired differences (Δ = Arm 3b − Arm 3a) are tested via two-sided Wilcoxon signed-rank tests ($\alpha = 0.05$), Hodges-Lehmann median estimators, and 95% subject-bootstrap confidence intervals (10,000 resamples).

VII. RESULTS AND DISCUSSION

A. Primary Hypothesis H_1 Outcome: Estimation Finding

In Protocol Amendment 3, the primary research question was framed as an estimation question: "*What is the difference in False Alerts per Hour between Arm 3b and Arm 3a* ($\Delta FA/hr = FA/hr_{Arm\ 3b} - FA/hr_{Arm\ 3a}$)*, with a 95% subject-bootstrap confidence interval?*" with stated targets ($\geq 60\%$ reduction in FA/hr relative to kinematic only gating and $\geq 85\%$ recall on genuine synthetic alert events preserved as stated targets, not as unflexible pass/fail criteria). In paired comparisons across the four evaluated conditions (Table IV, Fig. 2b), Arm 3b demonstrated no statistically detectable false-alert reduction over Arm 3a, with mean differences between -0.49 and $+0.13$ FA/h, all 95% CIs including zero, and Wilcoxon signed-rank p ranging from 0.28 to 0.98 (Table IV).

TABLE VI
LEAVE-ONE-SUBJECT-OUT COMBINED ALERTING BENCHMARK RESULTS ACROSS CONDITIONS (ALERT COUNTS ARCHIVED IN ALERT_COUNTS.CSV)

Condition	Alerting Arm	Recall (%)	FA/hr	Precision (%)	F_1 Score
Condition 0 (No-Fault)	Arm 1 (Fixed)	**91.1**	3.20	56.9	0.700
	Arm 2 (Personal)	80.4	5.69	42.8	0.558
	Arm 3a (+Kinematic)	80.4	1.47	74.3	0.772
	Arm 3b (+Trust Score)	79.5	1.69	71.0	0.750
	Arm 3b-noMotion	79.5	2.91	58.8	0.676
	Arm 4 (conformal)*	72.3	**0.70**	**83.7**	**0.776**
Regime A (Low Duty $\approx 10\%$)	Arm 1 (Fixed)	**91.1**	3.58	54.1	0.679
	Arm 2 (Personal)	80.4	6.08	41.0	0.543
	Arm 3a (+Kinematic)	80.4	1.63	72.1	**0.760**
	Arm 3b (+Trust Score)	79.5	1.82	69.4	0.741
	Arm 3b-noMotion	79.5	3.07	57.3	0.666
	Arm 4 (conformal)*	62.5	**0.67**	**82.1**	0.710
Regime B (Nominal $\approx$ 25–30%)	Arm 1 (Fixed)	**91.1**	4.51	48.4	0.632
	Arm 2 (Personal)	75.9	7.16	35.8	0.487
	Arm 3a (+Kinematic)	75.9	2.30	63.5	**0.691**
	Arm 3b (+Trust Score)	75.0	2.37	62.2	0.680
	Arm 3b-noMotion	75.0	3.61	51.9	0.614
	Arm 4 (conformal)*	55.4	**0.70**	**79.0**	0.651
F7 (High-Motion BVP Noise)	Arm 1 (Fixed)	**92.0**	3.07	57.7	**0.709**
	Arm 2 (Personal)	64.3	4.61	41.0	0.501
	Arm 3a (+Kinematic)	64.3	1.57	67.1	0.657
	Arm 3b (+Trust Score)	58.0	1.28	69.2	0.631
	Arm 3b-noMotion	58.0	2.05	58.4	0.582
	Arm 4 (conformal)*	54.5	**0.35**	**88.2**	0.673

**Pre-specified in Amendment 3, implemented after other arms' results were seen.*

B. Secondary Benchmark Findings

Kinematic motion gating drives alert accuracy: Kinematic stillness gating accounted for the largest share of false-alert suppression: relative to ungated Arm 2, Arm 3a reduced false alerts per hour by 74.2% in clean data (1.47 vs 5.69 FA/h) and by 67.9% in Regime B (2.30 vs 7.16 FA/h) (Table VI). Furthermore, the no-motion trust ablation (Arm 3b-noMotion) alone reduced clean false alerts by about half (5.69 to 2.91 FA/h).

Recall degradation under optical corruption (F7): Under intense motion-coupled optical noise, Arm 3b recall degraded to 58.0% compared to 64.3% in Arm 3a; two subjects drive the F7 recall difference (exploratory).

C. Edge-of-Grid Analysis and Fallback Rule Invocation

Hyperparameter tuning consistently selected parameter boundary values. Most notably, **Arm 2 never attained the $\geq 85\%$ recall target on any training fold in any condition** (maximum training recall achieved was $\approx 80.4\%$). Consequently, the fallback selection rule (maximizing training recall) was invoked in 100% (15/15) of folds for Arm 2 across all regimes, selecting $k = 2.0$ and $P = 10$ s. For Arm 1, tuning selected $\tau_{\text{pop}} = 100$ bpm and $P = 30$ s across all folds.

D. Arm 4 Conformal Abstention Evaluation (Pre-specified in Amendment 3, Implemented After Other Arms' Results Were Seen)

To benchmark distribution-free selective alerting, Arm 4 was pre-specified in Amendment 3, implemented after the other arms' results were seen. Inductive conformal calibration was performed on resting training windows with miscoverage parameter α tuned over $\{0.05, 0.10\}$ ($\alpha = 0.10$, $\lambda = 0.5$, $\theta_{\text{unc}} = 0.30$, and $P = 10$ s selected across all 15 LOSO training folds). Across all evaluated regimes, Arm 4 achieved the lowest clean false-alert frequency observed in the benchmark: 0.70 FA/h in clean data (a 52.4% reduction over Arm 3a and an 87.7% reduction over Arm 2), 0.67 FA/h in Regime A, and 0.35 FA/h under F7 optical noise (Table VI). Precision reached 83.7% in clean telemetry and 88.2% under F7 noise ($F_1 = 0.776$ clean, 0.673 F7) on the alert-count basis (Table VI); Arm 4's clean F_1 differs from Arm 3a (0.772) by 0.004 with no confidence interval and is not claimed as an improvement. However, conservative abstention came at the expense of sensitivity: episode recall dropped from 80.4% to 72.3% (81/112 episodes) in clean data, and to 54.5% (61/112 episodes) in F7 noise, demonstrating the fundamental trade-off between strict false-alert suppression and event sensitivity under non-stationary ambulatory noise.

VIII. LIMITATIONS

- **Synthetic Event Injection:** Resting tachycardia episodes were synthetically injected due to natural event sparsity. Arm 1 recall ($> 91\%$) depends on the $[105, 135]$ bpm injection range.
- **Cohort Homogeneity:** Evaluated in 15 healthy individuals during structured lab and ambulatory tasks; free-living tracking is not yet validated.
- **Error Propagation:** Front-end optical tracking inaccuracies propagate directly into downstream baseline and trust computation.
- **Subject S5 Outlier:** Subject S5 had chronic elevated resting pulse rate (reference ECG clean) and required dual pooled and stratified reporting (Table V).
- **Post-Hoc Conformal Implementation:** Arm 4 was pre-specified in Amendment 3 but implemented after other arms' outcomes were observed.
- **External Corpora:** Evaluation on external cohorts (PhysioNet Wearable, WESAD) was deferred to future work.

IX. FUTURE WORK

Future efforts will extend this benchmark along five technical avenues: (1) online dynamic conformal martingales under

TABLE VII
PROTOCOL AMENDMENTS AND PROVENANCE (FROM GIT LOG)

Commit	Date	Change
`41eff1e`	2026-10-18	Frozen protocol: H_1–H_3, faults F1–F7, LOSO CV.
`3a5ccc2`	2026-10-19	Enforced median norm check $[0.8, 1.2]$ g on ENMO.
`e358bf1`	2026-10-19	Adapted ramp rates clamped to $[105, 135]$ bpm.
`cad4c7f`	2026-10-19	StandardAlertEmitter ($P \in \{10, 20, 30\}$ s, $R = 60$ s).
`a9ce657`	2026-10-19	Widened search grids for τ_{pop} and k on boundaries.
`f92ca8d`	2026-10-19	Fallback rule invoked for Arm 2 in 100% of folds.
(no commit)	2026-10-22	Draft Amendment 5 (unratified, not committed): Arm 4 pre-specified in Amendment 3, implemented after other arms' results ($\alpha = 0.10$).

covariate shifts [4]; (2) external wearable corpora evaluation, including PhysioNet Wearable [6] and WESAD; (3) ambulatory inpatient holter validation with spontaneous arrhythmias; (4) on-device microcontroller deployment of causal baseline and kinematic gating engines; and (5) prospective clinical validation of operational symptom review (FR-06) and local explanation (FR-07) modules.

X. CONCLUSION

In this pre-specified and version-tagged fault-injection benchmark, kinematic motion gating accounted for the largest share of false-alert suppression in wearable heart-rate monitoring, while the no-motion trust ablation (Arm 3b-noMotion) alone reduced clean false alerts by about half (5.69 to 2.91 FA/h). Multi-component software trust scoring showed no statistically detectable incremental benefit over kinematic stillness gating. Highest recall but lower precision was reached with fixed thresholding with persistence. Recall is directly dependent on the injection range of 105–135 bpm. All findings are based on synthetic tachycardia episodes in healthy subjects and have no clinical relevance.

REGULATORY, DATA, AND CODE AVAILABILITY

Regulatory Status (Not a Medical Device): VitaSense AI is software, alerting algorithms and benchmark suite are academic research artifacts. They are not cleared or approved by any regulatory authority and are not intended for medical diagnosis, clinical treatment or life-critical monitoring.

Availability of data and code: PPG-DaLiA is publicly available at UCI [1], [5]; PhysioNet Wearable is licensed under ODC-By v1.0 [6]. Benchmark code, execution scripts and numerical provenance logs will be released with the final version. Alert counts are archived in `alert_counts.csv` (in the supplementary files).

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